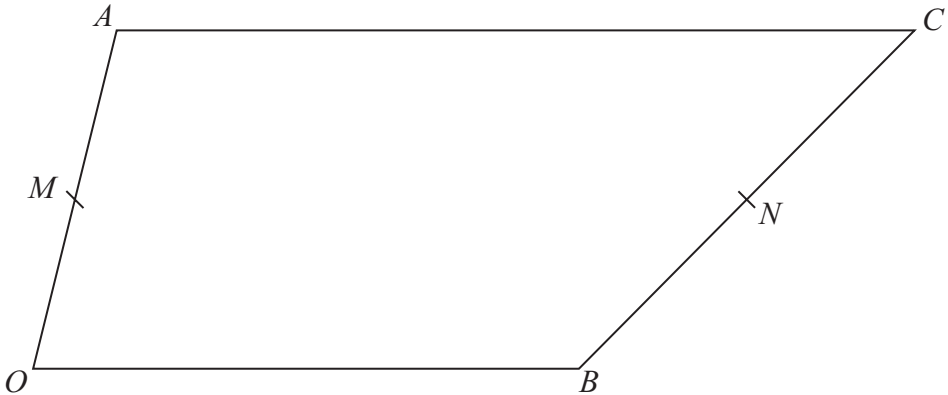


19 The diagram shows quadrilateral $OACB$.



M is the midpoint of OA .
 N is the point on BC such that $BN:NC = 4:5$

$\vec{OA} = \mathbf{a}$ $\vec{OB} = \mathbf{b}$ $\vec{AC} = k\mathbf{b}$ where k is a positive integer.

- (a) Express $\vec{MN}$ in terms of k , $\mathbf{a}$ and $\mathbf{b}$.
 Give your answer in its simplest form.

(4)

- (b) Is MN parallel to OB ?
 Give a reason for your answer.

(1)